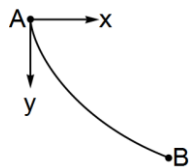


Chap. 26 Optics: The Principle of Least Time

1. The Curve of Fastest Descent (II): Optical Analog

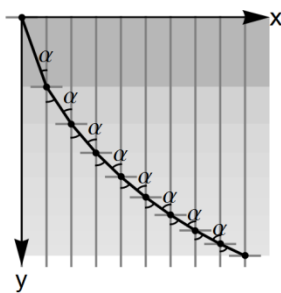
The brachistochrone problem (or the problem of fastest descent) was posed by Johann Bernoulli in *Acta Eruditorum* in June 1696: given two points A and B in a vertical plane, what is the curve on which a particle, acted on only by gravity, starts at A and reaches B in the shortest time. To solve this problem, Johann Bernoulli appeals to Fermat's principle, namely that light always follows the shortest possible time of travel.

What he considered is not a particle descending along a curve, but a beam of light travelling in a non-uniform medium. The index of refraction is such *designed* that the speed of light increases *as if* following a constant vertical acceleration (that of gravity g). It is not saying that the speed of light increases when propagating downwards in a gravity field in the same manner as the mass particle, but just to analog the effect of gravitational acceleration on a mass by using the refraction index gradient of the medium, and thus there is no gravitational effect for light.



(a) What should be the index of refraction $n(y)$ as a function of vertical position y in order to allow the speed of light in the medium to be just the same as a mass in the gravity field g . (Note that the y axis has been flipped down.)

According to Fermat's principle, the propagation of light automatically chooses the curve of the shortest travelling time, and hence the curve of fastest descent is simply the trajectory of light in the medium with index $n(y)$.



Johann Bernoulli noted that Fermat's principle is equivalent to the law of refraction. So he divides the plane into strips and he assumes that the particle follows a straight line in each strip. The path is then piecewise linear. The problem is to determine the angle α of the straight line segment in each strip as a function of y by the law of refraction.

(b) Show that the vertical position y and the angle α are related by $y = 2R \sin^2 \alpha$, where R is a constant of the motion.

(c) Use trigonometry to express α in terms of dy/dx , so as to obtain the differential equation that determines the curve of fastest descent.

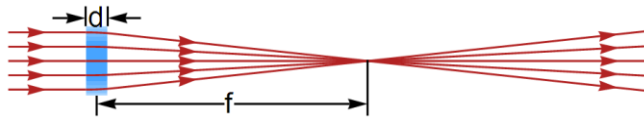
(d) Show that the inverted cycloid,

$$\begin{cases} x = R(\theta - \sin \theta) \\ y = R(\cos \theta - 1) \end{cases}$$

satisfies the differential equation, and is therefore the solution to the brachistochrone problem.

2. Gradient Index (GRIN) Lenses

Gradient index (GRIN) materials are materials that have non-uniform index of refraction. It can be fabricated by the hierarchically structured polymer optical materials. By layering polycarbonate (PC) ($n = 1.58$) and polymethylmethacrylate (PMMA) ($n = 1.49$) films, the index of refraction can be tuned between 1.49 and 1.58 smoothly throughout the material. The GRIN material show promise for use in optical systems that have fewer, lighter lenses than traditional lens systems. The following figure shows how light is focused by a flat GRIN lens, which is simply a polymer disk of uniform thickness $d = 0.5\text{cm}$, but with non-uniform index.



The index of refraction at r distance away from the center of the lens is described by the function

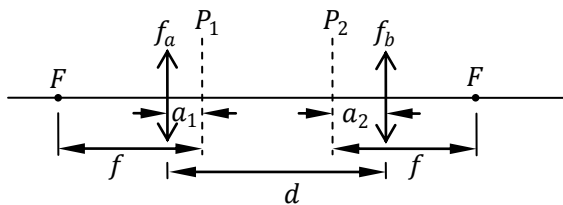
$$n(r) = n_0 - \alpha r^2,$$

where $n_0 = 1.58$ and $\alpha = 0.167\text{cm}^{-2}$. Use Fermat's principle to determine the focus f of the GRIN lens.

Chap. 27 Geometrical Optics

1. Principle Planes of Compound Lenses

In the Lecture, we have learnt that the optical system of compound lenses can be considered as a thin lens of focus f separated to the two principal planes that rays received at one principle plane is transferred to the corresponding point of the other. Here is an example for you. Two thin lenses with the focus lengths f_a and f_b are placed by the distance d . The principle planes P_1 and P_2 are located at a_1 behind the front lens and a_2 in front of the back lens respectively. Find the expressions for a_1 , a_2 and f in terms of f_a , f_b and d .

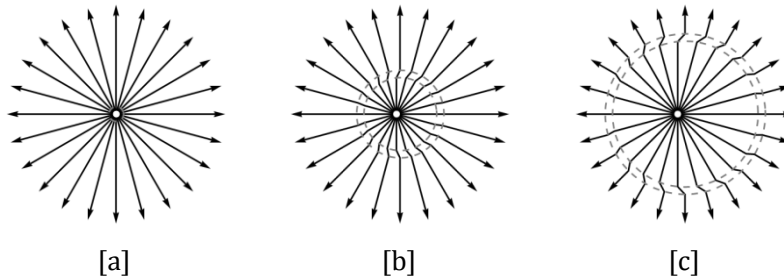


Chap. 28 Electromagnetic Radiation

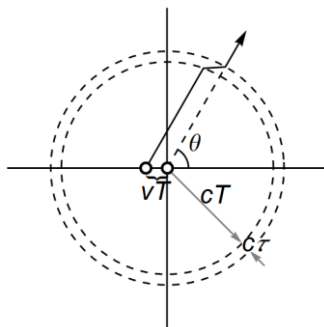
1. Radiation From an Accelerated Charge

We have learnt from the Lecture, that an accelerating charge gives out electromagnetic radiations. J. J. Thomson provides a simple derivation for the intensity and angular distribution of the radiation from a point charge subject to an arbitrary but small acceleration.

We start by taking a point charge q at rest at time $t = 0$. Coulomb's law implies that its electric field lines will be purely radial as shown in the figure [a] below. We now accelerate the charge (to the left) by the acceleration a for a small time interval τ , and look at the field line configuration at a time T after this. Then we will see a thin shell of kinks of the field lines propagates outwards with the speed of light, as in the figure [b] and [c].



Why is there a kink in the field line? Because outside of a sphere of radius $R = cT$, the news about the acceleration has not arrived yet, so that the field line still points towards the original position of the charge. While inside this sphere, the electric field has been told that the charge is now moving with a constant speed $v = a\tau$, so that the field lines are now able to “stick to” and “move with” the charge, as what it should look like the charge's frame. So there's an inner region with field lines pointing away from the charge, and an outer region with field lines pointing away from the original position of the charge. These two regions are separated by a transition region: a spherical shell of the thickness $c\tau$ that expands outward at the speed of light.



The next question is, what does the field look like in the transition region? The electric field lines have to be connected! Otherwise it will contradict to Gauss's Law. Therefore in the transition region, the electric field has a non-radial component E_θ , and the electric field can be written as $\mathbf{E} = E_r \mathbf{e}_r + E_\theta \mathbf{e}_\theta$, where E_r is the radial component.

- (a) What is the radial component E_r of the electric field at a distance R from the charge? (Assume $v \ll c$ in the non-relativistic limit.)
- (b) Determine the tangent component E_θ in the transition region (expressed as a function

of R). Hint: use the geometric relations between E_r and E_θ .

Note that E_θ falls off with distance as R^{-1} in contrast to $E_r \sim R^{-2}$. Thus far from the accelerated charge (large R), only the tangent component E_θ will contribute significantly to the radiation field. The energy per unit volume stored in this field, proportional to $|\mathbf{E}|^2$, therefore fall off as R^{-2} , so the total energy contained in the shell is unchanged as the shell expanded.

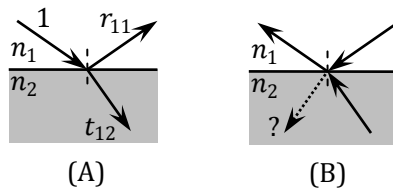
(c) According to Poynting's Law, determine the energy flow density $\mathbf{S}$ (the Poynting vector) in each direction θ . Hint: analog to the propagation of plane-wave in the free space.

(d) What is the total power of radiation?

Chap. 29 Interference

1. Interference on the Interface

Reflection and refraction happen on the interface of two mediums (1 and 2) with different indexes of refraction n_1 and n_2 . Suppose the incident light in medium 1 is coming with the amplitude 1 (as in Fig. A). By saying that the reflection coefficient from medium 1 to 1 is r_{11} , and the transmission coefficient from medium 1 to 2 is t_{12} , we mean that the amplitude of the reflected and refracted light will be r_{11} and t_{12} respectively.

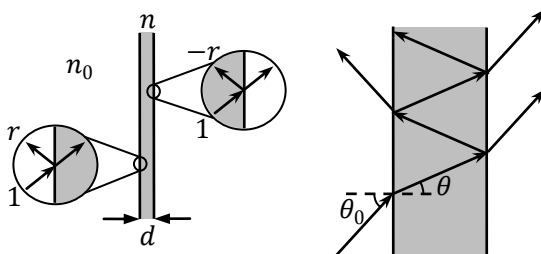


According to the principle of ray reversibility, if the reflected and refracted light proceed in the reverse direction along the same path, the incident light of the original amplitude should be generated (see Fig. B). One may doubt is an extra ray will be generated as the dashed ray in Fig. B. This extra ray contradicts with the reversibility principle, and is fairly possible if we consider light as particles. The wave nature of light saves the reversibility principle, which indicates that the light will undergo completely destructive interference on the path of the extra ray, and thus the extra ray does not exist.

If the above statement is true, what should be the relations between r_{11} , r_{22} , t_{12} and t_{21} ?

2. Film Interference

A thin film of the thickness d and refractive index n is sandwiched in the medium of the refractive index n_0 . The coefficient of reflection on the interface is r . Light shined on the film will be reflected between both interfaces for many times. Each time the light reaches the back interface produces a transmission wave. The transmission waves interfere with each other leads to the phenomenon known as the film interference.



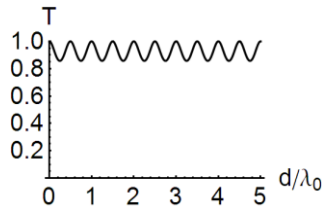
(a) Suppose the incident angle is θ_0 , and the wave length in the medium n_0 is λ_0 , show that the transmittance of the film takes the following form

$$T = \frac{1}{\alpha \sin^2 \delta + 1},$$

and determine the coefficient α and the angle δ .

The equal thickness interference is a special case of film interference, when the incident angle is fixed at zero.

(b) Fix $\theta_0 = 0$ and plot the transmittance as a function of film thickness d . Explain the oscillations you see.



Somehow by coating the surface, people are able to change the reflection coefficient r .

(c) How does the equal thickness interference affected by the coefficient r , especially when $r \rightarrow 1$? Find the standing wave modes in resonance with the incident light that leads to the peaks in transmittance.

The equal inclination interference is another case of film interference, in which the thickness is fixed but the incident angle varies. The equal inclination interference is more complicated because the coefficient of reflection also changes with the incident angle. Let us suppose hereon $n_0 = 1.5$ (glass) and $n = 1$ (air), and the incident light is polarized to S wave.

(d) For $d = 0.1n_0\lambda_0$, $d = n_0\lambda_0$ and $d = 5n_0\lambda_0$, plot the transmittance as a function of the incident angle θ_0 for θ_0 ranging from 0 to $\pi/2$. Explain the oscillations in the transmittance. Hint: you may determine how r depends on θ_0 first.

(e) Why does the transmittance drops to zero at around a critical angle $\theta_c = 41^\circ 49'$? What is the physical meaning of this angle?

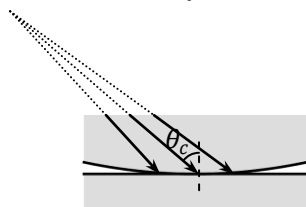
Finite transmittance remains when the incident angle exceeds the critical angle. This is a quantum mechanical effect of light, known as the optical tunneling, that can be demonstrated by simple experiments even in the undergraduate lab.

(f) Calculate the transmittance as a function of film thickness at critical angle $\theta_0 = \theta_c$.

Optical tunneling is prominent only for very thin films whose thickness should be controlled within a few wavelength of light. To realize a thin film like this, we evoke the Newton's ring device. Around the center of Newton's ring, there is a centimeter large region in which the air layer thickness varies smoothly within several wavelength according to

$$d = \frac{x^2 + y^2}{2R},$$

where $\mathbf{r} = (x, y)$ is the displacement from the center.

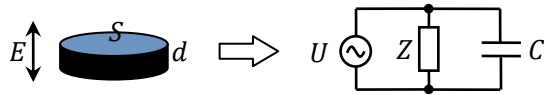


(g) What kind of transmission pattern do you expected to observe on the other side of Newton's ring device, if the light is emitted from a point source and falls on the center at the critical angle. Hint: design a computer program to help you calculate the transmitted light intensity at each point (x, y) .

Chap. 31 Origin of the Refractive Index

1. Plasma Oscillation (III): Absorption of Light

In the Problem 1 of Chap. 23 we have shown that a thin metal cylinder can be considered as a Z - C circuit effectively. When the electromagnetic wave is beamed on the metal, the oscillating electric field E applied across the metal will introduce a AC voltage source U in the effective circuit.



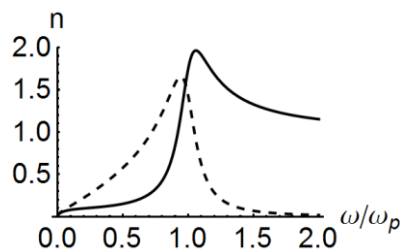
(a) Calculate the total current I in the metal (in terms of Z and C) in response to the electric field E .

According to Maxwell's theory of displacement current, the current inside the metal will not end up at the surfaces, but will be continued by the displacement current in electromagnetic field. The displacement current *density* (i.e. current per unit sectional area) by definition is the time derivative of electric displacement field D .

(b) Find the AC dielectric constant $\epsilon(\omega)$ as the ratio between D and E . Express your result in terms of the vacuum dielectric constant ϵ_0 , the plasma frequency ω_p and the relaxation time τ . Hint: you should substitute the expression for Z and C . If you forget the result in the Problem 1 of Chap. 23, do it once more.

(c) Find the index of refraction for the metal. Hint: the index of refraction can be obtained as the ration of the phase velocity and the light speed. The phase velocity of light in the metal can be extracted from the wave equation, which can be deduced from the Maxwell equations ($\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$ and $\nabla \times \mathbf{H} = \partial_t \mathbf{D}$).

The following figure shows the real (solid line) and imaginary (dashed line) part of the index of refraction as a function of frequency ω of light for the case of $\omega_p \tau = 5$.



(d) Explain why metal are usually opaque to light. Roughly upon which frequency does the metal become transparent?

It is known that light does not have a rest mass in vacuum. This is why light can propagate with the speed c and does not contradict with special relativiey. However this is not true for the light in the metal. Due to the strong scattering with electrons in the metal, light will be slowed down as if the photon acquires a rest mass. This effect can be shown in the limit that the relaxation time tends to infinity, which is known as the *perfect metal* limit, when the resistivity drops to zero and the metal becomes a perfect conductor (not superconductor).

(e) Show the dispersion relation of light in the perfect metal, and find the rest mass of photon.

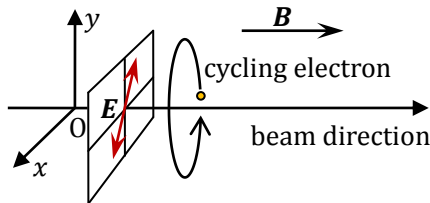
(f) As the photon acquires rest mass, it can be stopped. However the rest photon is never

observed in experiments. How does the nature forbid us to see the rest photon?

Chap. 33 Polarization

1. Faraday Effect

The Faraday effect or Faraday rotation is a magneto-optical phenomenon, where the direction of polarization is rotated about the beam axis as the light is traveling through the material with applied magnetic field in the parallel direction.



The Faraday effect is a result of ferromagnetic resonance of light and electrons in the magnetic field. Without the light, the electron in the material will be cycling in the magnetic field B with frequency ω_c .

(a) Determine the cycling frequency ω_c given the electron mass m and charge e .

When a beam of light falls on the electron, the electron will be driven back and forth in the cycling plane, hence undergoing a forced oscillation, which may go into resonance if the light frequency ω matches the cycling frequency ω_c .

(b) Write down the equation of motion for the electron.

The polarization of the material is related to the electron displacement by $\mathbf{P} = -nex$, where n is the electron density. The linear response of polarization to the electron field defines the *polarizability* χ , which in general is a matrix.

$$\mathbf{P} = \epsilon_0 \chi \mathbf{E}.$$

(c) Use the complex number method to find the polarizability matrix χ .

(d) Diagonalize the polarizability matrix to find all eigen modes with their eigen values respectively. Explain the physical meaning of the eigen modes.

The coupling of light to the cycling electron causes the linearly polarized ray to be decomposed into two circularly polarized rays which propagate at different speeds, a property known as circular birefringence.

(e) Find the index of refraction n for the right-handed and left-handed circularly polarized light.

The rays can be considered to re-combine upon emergence from the medium, however owing to the difference in propagation speed they do so with a net phase offset, resulting in a rotation of the angle of linear polarization.

(f) Determine the angle of Faraday rotation after light propagating for the distance d , up to the first order of the component of the applied magnetic field in the direction of the beam of light.

2. Coherency Matrix of Light Polarization (Optional)

In the Lecture, we have learn that the polarization state of light may be expressed by a

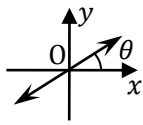
complex vector

$$|E\rangle = \begin{bmatrix} E_x \\ E_y \end{bmatrix},$$

which contains two complex numbers E_x and E_y that provides both the amplitudes and the phases of electric field vibrations in both x and y directions. For example, the polarization state pictured in [Fig. 33-2] are

picture	a	b	c	...	g	...
state $ E\rangle$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ e^{i\pi/4} \end{bmatrix}$	$\begin{bmatrix} 1 \\ i \end{bmatrix}$	...	$\begin{bmatrix} 1 \\ -i \end{bmatrix}$	...
polarization	linear	elliptic	circular (LH)	...	circular (RH)	...

(a) According to this notation, find the vector $|\theta\rangle$ that represents a linearly polarized state, with the polarization direction at angle θ with the x -axis.



In quantum mechanics, such a vector $|E\rangle$ is known as the state vector. Every state vector $|E\rangle$ has a dual, obtained by transposing and taking complex conjugate, known as the its adjoint vector, denoted by

$$\langle E| = [E_x^* \quad E_y^*].$$

The principle of quantum mechanics is that each state can be represented by the state vector. Physical observables are represented by operators (which means matrixes here). Each measurement can be taken by applying the corresponding matrix of that physical observable to the state vector, and then project to its adjoint vector, the resulting number is the expectation value of the physical observable.

For example, the light intensity is represented by the light intensity operator I , which is a 2×2 matrix

$$I = \begin{bmatrix} I_{xx} & I_{xy} \\ I_{yx} & I_{yy} \end{bmatrix}.$$

Then to measure the light intensity of the polarization state $|E\rangle$, we take the matrix I , apply to vector $|E\rangle$, and then project to the vector $\langle E|$.

$$\langle I \rangle = \langle E|I|E\rangle = [E_x^* \quad E_y^*] \begin{bmatrix} I_{xx} & I_{xy} \\ I_{yx} & I_{yy} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix}.$$

(b) Given that the light intensity is given by $\langle I \rangle = |E_x|^2 + |E_y|^2$, what should the matrix I be so as to produce this result.

(c) Find the matrix $I(\theta)$ that corresponds to the measurement of light intensity through a polarizer whose passing direction is at angle θ with the x axis.

Hints: You can try the measurement on different states, e.g. the x -polarized, y -polarized and θ -direction linearly polarized states, or circularly polarized states as well. From these special examples, you will be able to determine the four matrix elements of $I(\theta)$.

We have considered linearly, circularly and elliptically polarized light, which only covers a very small portion of the polarization states known as the *fully polarized* states. There is the case of *unpolarized* light, also called the *natural light*. Sun light is a typical example of natural

light. There are even more cases of *partially polarized* light, which are mixtures of fully polarized and unpolarized light.

Natural light does not have a certain polarization. It is a statistically combining of all polarization states with equal probability. Naively one may wish to describe the natural light by a superposition of all fully polarized states $|E\rangle$.

$$|N\rangle = \sum_E |E\rangle = 0.$$

But unfortunately the result is 0 due to the phase cancelation effect.

(d) Show that at least the superposition of all linearly polarized states results in zero. Hints: use integral instead of summation $\int d\theta |\theta\rangle$.

In quantum mechanics, the fully polarized states are called pure states, and all the other partially polarized and unpolarized states are called mixed states. Only pure states can be described by state vectors. Superposition of any pure states is still a pure state. Because the superposition of state vectors is a quantum mechanical mixing that has interference effect. To obtain the mixed state, statistical (thermal) mixing is required, which means that the states are mixed by probability but not by probability amplitude. Therefore there is no hope to represent mixed states by state vectors.

Nevertheless, both pure and mixed states can be represented by matrixes, known as the *coherency matrix*. The coherency matrix of light polarization is a 2×2 matrix. For fully polarized state $|E\rangle$, the coherency matrix ρ is defined as

$$\rho = |E\rangle\langle E|.$$

(e) Find the coherency matrix $\rho(\theta)$ representing the θ -direction linearly polarized light.

(f) Find the coherency matrix $\rho(R)$ and $\rho(L)$ representing the right-handed and left-handed circular light.

The coherency matrix of natural light can be obtained by combining the coherency matrix of all θ -direction linearly polarized light with equal probability.

(g) Find the coherency matrix of the natural light.

In the coherency matrix formalism, the measurement of a physical observable represented by matrix I is given by taking trace

$$\langle I \rangle = \text{Tr } \rho I.$$

(h) Calculate the angular distribution of light intensity for the θ -direction linearly polarized state. Hints: Use $I(\theta)$ to implement the measurement.

(i) Calculate the angular distribution of light intensity for the circularly polarized light and the natural light. Is there any difference between them?

(j) Find the determinant $\det \rho$ for both the circularly polarized light and the natural light. Can you see the difference?

A partially polarized light is in general described by the coherency matrix

$$\rho = \begin{bmatrix} a & c^* \\ c & b \end{bmatrix}.$$

Any partially polarized light can be described as a statistical mixing of a natural light with a fully polarized light. The degree of polarization (DOP) is a quantity used to describe the portion of light which is polarized, defined as the ratio of the light intensity of polarization component to the total light intensity. The fully polarized light has a DOP equal to 1, while the DOP for natural light is 0.

(k) Determine the DOP given the coherency matrix ρ . Under what condition does the polarization lose coherence completely and become natural light?

Hints: decompose ρ into the summation of fully polarized and unpolarized parts. The fully polarized coherency matrix has zero determinant.

The effect of a polarizer is a projection operator $P(\theta)$ which is also a 2×2 matrix. With the initial polarization described by the coherency matrix ρ , the light is sent into a polarizer whose pass axis is at angle θ . The polarization of the out-coming light will be described by $P(\theta)\rho P^\dagger(\theta)$.

(l) Find the matrix $P(\theta)$.

The birefringence effect of cellophane can be formulated in similar way. Each cellophane is represented by an unitary matrix U . Light passing through a cellophane changes the polarization state by $SU(2)$ rotation of the coherency matrix $\rho \rightarrow U\rho U^\dagger$. For example, the matrix representation of a quarter-wave plate is

$$U = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}.$$

It is easy to show that the quarter-wave plate changes the linearly polarized light in [Fig. 33-2 a] to circular polarization.

If light passes through a series of polarizers P and cellophanes U , the effect is to multiply the P or U matrixes together one by one in right order. Then perform the transform on the coherency matrix of incident light results in the coherency matrix of the transmitted light.

(m) Use the matrix representation of polarizer to demonstrate the paradox that some light will be transmitted if a third polarizer with pass axis at 45° is plugged in between two polarizers of perpendicular pass axes.